Aakash
Medical | IIT-JEE | Foundations
(Divisions of Aakash Educational Services Limited)

24/02/2021 **CoE-XII**

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NCERT BOOSTER TEST SERIES

Test – 08

Topics covered in various subjects:

Physics : Electromagnetic Induction, Alternating Current, Electromagnetic Wave

Chemistry : The p-Block Elements, The d and f-Block Elements, Coordination Compounds

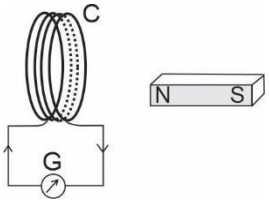
Botany : Biodiversity and Conservation, Environmental Issues

Zoology : Complete Syllabus of XI and XII

PHYSICS

Choose the correct answer:

1. On the basis of figure as shown, choose the correct statement



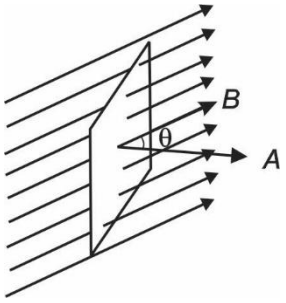
(1) If relative motion between the coil and magnet is zero, then no current will be induced in the coil

(2) If relative motion between the coil and magnet is non-zero, then current will be induced in the coil

(3) The deflection (and hence current) is found to be larger when the magnet is pushed towards or pulled away from the coil faster

(4) All of these

2. If a plane of area A placed in a uniform magnetic field B shown in figure, then magnetic flux passing through the area will be



(1) $\vec{B} \times \vec{A}$

(2) $\vec{A} \cdot \vec{B}$

(3) $AB\tan\theta$

(4) None of these

3. A square loop of side 5 cm and resistance 1Ω is placed vertically in the east-west plane. A

uniform magnetic field of 0.20 T is set-up across the plane in the north-east direction. The magnetic field is decreased to zero in 0.20s at a steady rate, the magnitude of induced current during this time-interval will be

(1) $\frac{2.5}{\sqrt{2}}$ mA (2) $\frac{25}{\sqrt{2}}$ mA

(3) 20 mA (4) $10\sqrt{2}$ mA

4. A circular coil of radius 20 cm, and resistance 4Ω is placed with its plane perpendicular to the horizontal component of earth's magnetic field. It is rotated about its vertical diameter through 180° in 0.5 s. The magnitude of emf induced in the coil will be ($B_H = 3.0 \times 10^{-5}$ T)

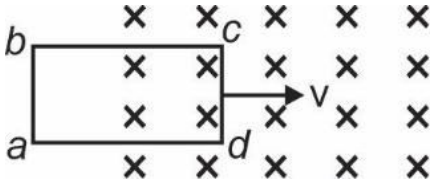
(1) 2.4 πμV

(2) 3.6 πμV

(3) 1.2 πμV

(4) 4.8 πμV

5. A rectangular loop is moving into a region of uniform magnetic field which is directed normal to the plane of the loop as shown in figure, choose the correct statement



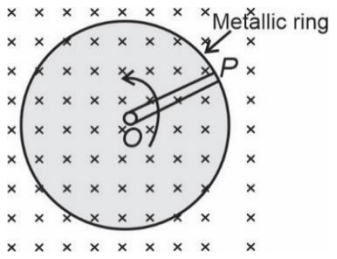
(1) Magnetic flux through the loop increases

(2) Induced current in the loop will be anticlockwise

(3) After sometime, current in the loop may be zero

(4) All of these

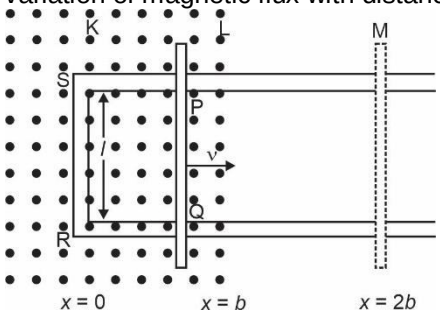
6. A metallic rod of 2 m length is rotated with a frequency of 25 rev/s, with one end hinged at the centre and other end at the circumference of a circular metallic ring of radius 2 m, about an axis passing through the centre and perpendicular to the plane of the ring as shown in figure. A constant and uniform magnetic field of 2*T* parallel to the axis is present everywhere. The emf between the centre and metallic ring will be



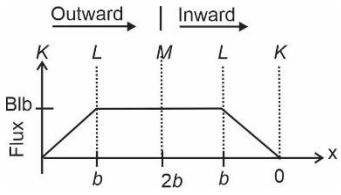
(1) 314 V (2) 628 V

(3) 168 V (4) 268 V

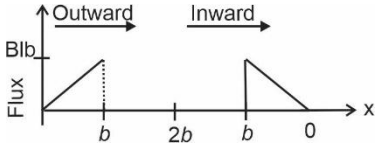
7. The arm PQ of the rectangular conductor is moved from x = 0, outwards. The uniform magnetic field is perpendicular to the plane and extends from x = 0 to x = *b* and is zero for x > *b*. Only the arm PQ is pulled outwards from x = 0 to x = 2*b* and is then moved back to x = 0 with constant speed. Choose the correct variation of magnetic flux with distance.



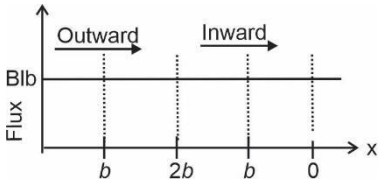
(1)



(2)



(3)



(4) None of these

8. Which among the following is the application of eddy current?

(1) Electric power meters

(2) Induction furnace

(3) Moving coil galvanometer

(4) Both (1) & (2)

9. The dimensional formula of the mutual inductance is

(1) $[ML^2T^{-2}A^{-2}]$ (2) $[ML^2T^{-3}A^{-1}]$

(3) $[ML^2T^{-2}A^{-1}]$ (4) $[ML^2T^{-1}A^{-1}]$

10. The magnetic energy stored per unit volume in a solenoid of magnetic field B, area A and length l, is

(1) $\frac{B^2Al}{\mu_0}$ (2) $\frac{B^2}{2\mu_0}$

(3) $\frac{B^2}{\mu_0}$ (4) $\frac{B^2Al}{2\mu_0}$

11. A circular coil of radius 4 cm and 10 turns is rotated about its vertical diameter with an angular speed 50 rad/s in a uniform horizontal magnetic field of magnitude 10^{-2}T. The maximum emf induced in the coil is nearly

(1) 1.2×10^{-2} V (2) 2.5×10^{-2} V

(3) 1.6×10^{-2} V (4) 4.2×10^{-2} V

12. A long solenoid with 10 turns per cm has a small loop of area 4.0 cm^2 placed inside the solenoid normal to its axis. If the current carried by the solenoid changes steadily from 1.0 A to 3.0 A in 0.2 s. The induced emf in the loop while the current is changing, is nearly

(1) 10 μV (2) 2.5 μV

(3) 5 μV (4) Zero

13. A rectangular wire loop of sides 10 cm and 5 cm with a small cut is moving out of a region of uniform magnetic field of 0.5 T directed normal to the loop, the emf developed across the cut if the velocity of the loop is 2 cm/s in a direction normal to the longer side, is

(1) 1 mV which lasts for 2.5 s

(2) 1 mV which lasts for 5 s

(3) 0.5 mV which lasts for 2.5 s

(4) 2 mV which lasts for 5 s

14. Current in a circuit falls from 6.0 A to 0.0 A in 0.2 s. If an average emf of 600 V induced. The estimated self inductance of the coil will be

(1) 10 H (2) 20 H

(3) 5 H (4) 28 H

15. A pair of adjacent coils has a mutual inductance of 3H. If the current in one coil changes from 0 to 15 A in 0.1 s. The change of flux linkage with the other coil will be

(1) 45 Wb (2) 60 Wb

(3) 30 Wb (4) 20 Wb

16. **Statement (A)**: The self-induced emf opposes any change (increase or decrease) of current in the coil.

Statement (B): The self-inductance of the coil depends on its geometry and on the permeability of the medium.

(1) Only statement (A) is correct

(2) Only statement (B) is correct

(3) Both the statements (A) and (B) are correct

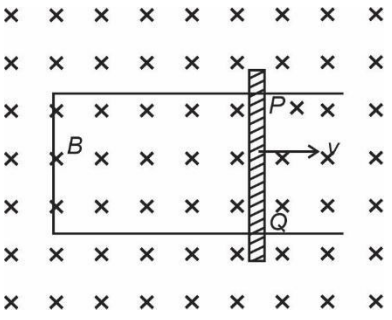
(4) Neither statement (A) nor (B) is correct

17. Two concentric circular coils, one of large radius r_1 and other of small radius r_2, such that $r_1 >> r_2$, are placed co-axially with centres coinciding. The mutual inductance of the arrangement will be

(1) $\frac{\mu_0\pi r_2^2}{2r_1}$ (2) $\frac{\mu_0\pi r_1^2}{2r_2}$

(3) $\frac{\mu_0\pi r_2}{2r_1}$ (4) $\frac{\mu_0\pi r_1}{2r_2}$

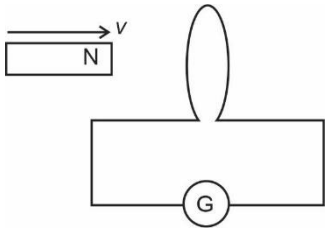
18. A conducting rod PQ of length l moving with constant velocity v on parallel rails in the uniform perpendicular magnetic field B as shown in the figure. The force required to move the rod with constant velocity v is proportional to



(1) v (2) $v^{\frac{1}{2}}$

(3) v^2 (4) v^{-1}

19. If a bar magnet moving towards the coil crosses the coil, then current in the coil



(1) Increases

(2) Decreases

(3) First increases then decreases

(4) First decreases then increases

20. An ac voltage is applied across a resistor of resistance *R*. The phase difference between the current and voltage will be

(1) $\frac{\pi}{2}$ (2) π

(3) $\frac{\pi}{4}$ (4) Zero

21. A light bulb is rated at 60 W for a 200 V supply. The rms current through the bulb will be

(1) 0.6 A (2) 0.3 A

(3) 0.2 A (4) 0.4 A

22. An ac voltage $\varepsilon = \varepsilon_0 \sin \omega t$ is applied across the inductor of inductance *L*. The expression of current in the circuit will be

(1) $i = \frac{\varepsilon_0}{\omega L} \sin\left(\omega t + \frac{\pi}{2}\right)$

(2) $i = \frac{\varepsilon_0}{\omega L} \sin\left(\omega t - \frac{\pi}{2}\right)$

(3) $i = \varepsilon_0\ \omega L \sin\left(\omega t + \frac{\pi}{2}\right)$

(4) $i = \varepsilon_0\ \omega L \sin\left(\omega t - \frac{\pi}{2}\right)$

23. A pure inductor of inductance 20.0 mH is connected to a source of 220 V. If the frequency of the source is 60 Hz, then rms current in the circuit will be nearly

(1) 20 A (2) 29 A

(3) 9 A (4) 35 A

24. A voltage source $V = V_0 \sin \omega t$ is applied across a capacitor of capacitance *C*. The instantaneous power supplied to the capacitor will be

(1) $\frac{V_0^2}{\omega C} \sin \omega t$ (2) $\omega C V_0^2 \sin \omega t$

(3) $\frac{V_0^2}{2\omega C} \sin 2\omega t$ (4) $\frac{\omega C V_0^2}{2} \sin 2\omega t$

25. A 25.0 μF capacitor is connected to a 220 V, 60 Hz source. The capacitive reactance of the circuit will be

(1) 106 Ω (2) 121.6 Ω

(3) 144.5 Ω (4) 112.5 Ω

26. A resistor of 100 Ω and a capacitor of 20.0 μF are connected in series to 220 V, 50 Hz ac source. The voltage (r.m.s.) across the resistor will be

(1) 117 V (2) 63 V

(3) 92 V (4) 75 V

27. Which among the following electrical component dissipiate maximum power in ac circuit?

(1) Capacitor (2) Inductor

(3) Resistor (4) Both (1) and (2)

28. A sinusoidal voltage of peak value of 283 V and frequency 50 Hz is applied to a series LCR circuit in which $R = 3\ \Omega$, $L = 25.48$ mH and $C = 796\ \mu$F, The power factor of the circuit will be

(1) $\frac{3}{5}$ (2) $\frac{2}{5}$

(3) $\frac{3}{4}$ (4) $\frac{2}{3}$

29. Which among the following quantity is dimensionless?

(1) Quality factor (2) Reactance

(3) Power factor (4) Both (1) and (3)

30. An inductor of inductance L = 3.0 H, capacitor of capacitance C = 27μF and a resistor of resistance 15 Ω are connected in series LCR circuit. The resonance frequency (ω_r) of the circuit will be nearly

(1) 58 rad/s (2) 111 rad/s

(3) 29 rad/s (4) 88 rad/s

31. A 200 Ω resistor is connected to a 220 V, 50 Hz ac supply. The net power consumed over a full cycle will be

(1) 484 W (2) 242 W

(3) 154 W (4) 308 W

32. A power transmission line feeds input power at 2200 V to a step-down transformer with its primary windings having 2000 turns. If output power is received at 220 V, then the number of turns in the secondary coil of the transformer will be

(1) 200 (2) 400

(3) 100 (4) 800

33. An LC circuit contains a 20 mH inductor and a 50 μF capacitor with an initial charge of 5 mC. The resistance of the circuit is negligible. Let

the instant the circuit is closed be t = 0. The initial total energy stored in the circuit is

(1) 0.25 J (2) 1 J

(3) 0.6 J (4) 0.31 J

34. A circuit containing a 60 mH inductor and 40 μF capacitor in series connected to 220 V, 60 Hz supply. If resistance of the circuit is negligible, then average power transferred to inductor will be

(1) 40 W (2) 23.5 W

(3) 70.5 W (4) Zero

35. In the free oscillations of LC circuit, the sum of energies stored in the capacitor and inductor is (i_0 and q_0 are the maximum value of current and charge)

(1) $\frac{q_0^2}{2C}$ (2) $\frac{1}{2} L\, i_0^2$

(3) $\frac{q_0^2}{C}$ (4) Both (1) & (2)

36. Which among the following equation represents the Ampere-Maxwell law?

(1) $\oint \vec{E}.d\vec{l} = -\frac{d\phi_B}{dt}$

(2) $\oint \vec{B}.d\vec{l} = \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}$

(3) $\oint \vec{B}.d\vec{l} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}$

(4) $\oint \vec{E}.d\vec{A} = \frac{Q}{\varepsilon_0}$

37. In the propagation of electromagnetic wave, the electric field *E* and magnetic field B will be

(1) In opposite direction to each other

(2) In the same direction to each other

(3) Perpendicular to each other

(4) None of these

38. If the total energy transferred to a surface in time *t is U.* The magnitude of total momentum delivered to this surface (for complete reflection) is (symbol have their usual meaning)

(1) $\frac{U}{2c}$ (2) $\frac{2U}{c}$

(3) $\frac{U}{c}$ (4) $\frac{U}{4c}$

39. A plane electromagnetic of frequency 3 MHz travels in free space along the X-direction. At a particular point in space and time, $\vec{E} = 9.6\ \hat{j}$ V/m. The magnitude of magnetic field *B* at that point will be

(1) 2.1×10^{-8} T (2) 4.2×10^{-8} T

(3) 3.2×10^{-8} T (4) 3×10^{-8} T

40. The magnetic field in a plane electromagnetic wave is given by $B_y = 3 \times 10^{-8} \sin (2 \times 10^3\, x + 3 \times 10^{10}\, t)$T. The expression for the electric field will be

(1) $E_z = 9 \sin (2 \times 10^3 x + 3 \times 10^{10}\, t)$ V/m

(2) $E_z = 9 \sin (2 \times 10^3 x - 3 \times 10^{10}\, t)$ V/m

(3) $E_z = \sin (2 \times 10^3 x + 3 \times 10^{10}\, t)$ V/m

(4) $E_z = \sin (2 \times 10^3 x - 3 \times 10^{10}\, t)$ V/m

41. The maximum value of electric field produced by the radiation coming from a 50 W bulb at a distance of 3 m is (Assume that the efficiency of the bulb is 5% and it is a point source)

(1) 4.07 V/m

(2) 2.07 V/m

(3) 1 V/m

(4) 5 V/m

42. The microwave is detected by

(1) The eye photocells

(2) Thermopiles bolometer

(3) Point contact diodes

(4) Receiver's aerials

43. In a plane electromagnetic wave, the electric field oscillates sinusoidally at a frequency of 6.0×10^{10} Hz and amplitude 50 V/m, the wavelength of the wave will be

(1) 50 mm (2) 100 mm

(3) 10 mm (4) 5 mm

44. A capacitor made of two circular plates each of radius 8 cm, and separated by 4.0 cm. The capacitor is being charged by an external source. If the charging current is constant and equal to 0.3 A, then the rate of change of potential difference between the plates will be nearly

(1) 6.7×10^{10} V/s

(2) 4×10^9 V/s

(3) 6×10^{10} V/s

(4) 1×10^{10} V/s

45. Choose the correct statement among the following

(1) Long distance radio broadcast use short-wave bands

(2) It is necessary to use satellites for long distance TV transmission

(3) The small ozone layer on top of the stratosphere is crucial for human survival

(4) All of these

CHEMISTRY

46. The chemical formula of Indian saltpetre is

(1) $NaNO_3$ (2) KNO_3

(3) HNO_3 (4) $Ca(NO_3)_2$

47. Which one the following oxides is expected to have odd number of valence electron?

(1) N_2O_4 (2) NO_2

(3) SO_2 (4) N_2O_5

48. The correct order of increasing bond angles in the following species is

(1) $SbH_3 < AsH_3 < PH_3 < NH_3$

(2) $NH_3 < PH_3 < AsH_3 < SbH_3$

(3) $SbH_3 < AsH_3 < NH_3 < PH_3$

(4) $AsH_3 < SbH_3 < PH_3 < NH_3$

49. Which of the following order for the indicated property is wrong?

(1) $H_2O < H_2S < H_2Te$ - Acidic strength

(2) $Se < S < O < N$ - 1st ionisation potential

(3) $H_2O < NH_3 < PH_3$ - Basic strength

(4) $F^- < O^{2-} < N^{3-}$ - Ionic radius

50. Which of the following does not give nitrogen gas on heating?

(1) $(NH_4)_2Cr_2O_7$ (2) $Ba(N_3)_2$

(3) NaN_3 (4) NH_4Cl

51. When zinc is heated with dilute HNO_3 it produces

(1) $Zn(NO_3)_2$ and N_2O

(2) $Zn(NO_3)_2$ and NO_2

(3) $Zn(NO_3)_2$ and NO

(4) $Zn(OH)_2$ and $Zn(NO_3)_2$

52. When white phosphorous reacts with SO_2Cl_2 then the products formed are?

(1) $PCl_3 + SO_3$ (2) $PCl_5 + SO_2$

(3) $POCl_3 + SO_2$ (4) $PCl_3 + SO_2 + S_2Cl_2$

53. $NaOH + Cl_2$ (cold and dilute) gives

(1) NaCl + NaOCl

(2) $NaCl + NaClO_3$

(3) $NaCl + NaClO_4$

(4) $O_2 + NaClO_2$

54. When iodine reacts with chlorine in the presence of water it gives

(1) $HIO_4 + HCl$ (2) HOI + HCl

(3) $HClO_4 + HI$ (4) $HIO_3 + HCl$

55. Which of the following statements is correct for aqua regia?

(1) When three parts of conc. HCl and one part of conc. HNO_3 are mixed, aqua regia is formed

(2) When three parts of conc. HCl and one part of conc. HNO_2 are mixed, aqua regia is formed

(3) When three parts of conc. H_2SO_4 and one part of conc. HCl are mixed, aqua regia is formed

(4) When three parts of conc. HNO_3 and one part of conc. HCl are mixed, aqua regia is formed

56. Among the following, the correct order of electron gain enthalpy is

(1) He < Ne < Ar < Xe (2) He < Ar < Xe < Ne

(3) He < Xe < Ar < Ne (4) Xe < Ar < Ne < He

57. Which of the statements given below is incorrect?

(1) The noble gases except helium ($1s^2$) have completely filled ns^2np^6 electronic configuration in their valence shell

(2) Noble gases have high ionisation energy

(3) Noble gases are highly reactive in nature

(4) Noble gases have positive electron gain enthalpy

58. Which compound has distorted octahedral structure?

(1) XeF_2

(2) XeF_6

(3) XeO_2F_2

(4) XeO_3

59. Identify the incorrect statement regrading the molecule $XeOF_4$

(1) It is square pyramidal

(2) Xe is sp^3d hybridised

(3) There is one $p\pi$–$d\pi$ bond

(4) One lone pair present on central atom

60. Xe can be obtained by which of the following reaction?

(1) $XeF_4 + H_2O \rightarrow$

(2) $XeF_6 + H_2O \rightarrow$

(3) $XeF_2 + PF_5 \rightarrow$

(4) $XeF_6 + NaF \rightarrow$

61. When uranium is treated with ClF_3 it gives

(1) $UF_6 + ClF_5$ (2) $UF_4 + ClF$

(3) $UF_6 + ClF$ (4) $UF_4 + ClF_5$

62. Pick out the correct statement with respect to $[Ni(CN)_4]^{2-}$

(1) It is sp^3 hybridised and tetrahedral

(2) It is sp^3 hybridised and square planar

(3) It is dsp^2 hybridised and square planar

(4) It is dsp^2 hybridised and tetrahedral

63. The correct decreasing order of field strength of ligand is

(1) $CO > en > CN^- > NH_3$

(2) $CO > CN^- > en > NH_3$

(3) $en > CO > CN^- > NH_3$

(4) $NH_3 > en > CN^- > CO$

64. The IUPAC name of complex compound $K_2[PdCl_4)$ is

(1) Potassium tetrachloridopalladate (II)

(2) Potassium tetrachloridopalladium (II)

(3) Potassium tetrachloridopalladate (IV)

(4) Potassium tetrachloridopalladium (IV)

65. Number of possible geometrical isomers for the complex $[Co(NH_3)_4Cl_2]^+$ will be

(1) 4 (2) 3

(3) 2 (4) 6

66. The complexes $[Co(NH_3)_5(NO_2)]Cl_2$ and $[Co(NH_3)_5(ONO)]Cl_2$ are examples of which type of isomerism?

(1) Coordination isomerism

(2) Ionisation isomerism

(3) Solvate isomerism

(4) Linkage isomerism

67. If the spin only magnetic moment of $[MnBr_4]^{2-}$ is 5.9 B.M then predict the geometry of the complex ion

(1) Trigonal pyramidal

(2) Tetrahedral

(3) Square planar

(4) Octahedral

68. Which one of the following is/are expected to exhibit optical isomerism?

(ox = Oxalate) (en = ethylene diamine)

(1) $trans[CrCl_2(ox)_2]^{3-}$

(2) $cis[CrCl_2(ox)_2]^{3-}$

(3) $trans[Co(en)_2Cl_2]^+$

(4) Both (1) and (3)

69. Geometry of $[Cr(CO)_6]$ is

(1) Octahedral

(2) Square pyramidal

(3) Tetrahedral

(4) Trigonal bipyramidal

70. The coordination number and oxidation state of Cr in $K[Cr(H_2O)_2(C_2O_4)_2]$ are

(1) 4 and +3 (2) 3 and +3

(3) 6 and +2 (4) 6 and +3

71. Which of the following ligands is expected to be ambidentate?

(1) CH_2–NH_2 (2) H_2O

(3) SCN^- (4) $C_2O_4^{2-}$

72. Which of the following complexes is used for hydrogenation of alkene?

(1) $[Ni(CO)_4]$ (2) $[(Ph_3P)_3RhCl]$

(3) $Cis[PtCl_3(NH_3)_2]$ (4) $[Ag(CN)_2]^-$

73. The correct order of the stoichiometries of AgCl formed when $AgNO_3$ in excess is treated with complexes: $CoCl_3.5NH_3$, $CoCl_3.4NH3$, $CoCl3.6NH_3$ respectively is

(1) 2AgCl, 1AgCl, 3AgCl

(2) 3AgCl, 1AgCl, 2AgCl

(3) 2AgCl, 3AgCl, 1AgCl

(4) 1AgCl, 2AgCl, 3AgCl

74. Which of the following statements about $[Co(NH_3)_6]^{3+}$ is true

(1) It has two unpaired electrons and will be in high spin configuration

(2) It has no unpaired electrons and will be in low spin configuration

(3) It has five unpaired electrons and will be in high spin configuration

(4) It has four unpaired electrons and will be in high spin configuration

75. Magnetic moment 3.87 BM is given by (Atomic number : Mn = 25, V = 23, Cr = 24, Co = 27)

(1) Cr^{2+} (2) Mn^{2+}

(3) Co^{3+} (4) V^{2+}

76. The reaction of aqueous $KMnO_4$ with iodide in neutral medium gives

(1) Mn^{2+} and I_2 (2) Mn^{2+} and IO_3^-

(3) MnO_2 and I_2 (4) MnO_2 and IO_3^-

77. Which ion is not stable in aqueous solution?

(1) Mn^{2+} (2) Co^{3+}

(3) Cu^{+} (4) Fe^{3+}

78. $KMnO_4$ on reaction with oxalate ion in acidic medium changes to

(1) MnO_2 (2) Mn^{3+}

(3) Mn^{2+} (4) MnO_4^{2-}

79. Constituents of mischmetal is

(1) Ln and Fe (2) La and C

(3) La and Cu (4) Fe and Ca

80. Correct statement about dichromate ion is

(1) 6Cr–O bond lengths are equal

(2) Bond angle of Cr–O–Cr is 109.5°

(3) All Cr–O bond lengths are equal

(4) It is pink in colour in aqueous medium

81. Which of the following statements about the interstitial compounds is correct?

(1) They are chemically reactive in nature

(2) They retain metallic conductivity

(3) They have low melting points, lower than those pure metals

(4) They are very soft

82. Mn^{2+} reacts with peroxodisulphate according to the equation

$2Mn^{2+} + 5S_2O_8^{2-} + 8H_2O \rightarrow 2MnO_4^- + 10SO_4^{2-} + 16H^+$ Here 10 mL of 0.1M Mn^{2+} is equivalent to

(1) 50 mL of 0.1M $S_2O_8^{2-}$

(2) 25 mL of 1M $S_2O_8^{2-}$

(3) 25 mL of 0.1M $S_2O_8^{2-}$

(4) 20 mL of 0.5M $S_2O_8^{2-}$

83. The electronic configuration of Cm (Z = 96) is

(1) $[Rn]_{86}5f^7 6d^1 7s^1$ (2) $[Rn]_{86}5f^7 7s^2$

(3) $[Rn]_{86}5f^7 6d^1 7s^2$ (4) $[Rn]_{86}6d^2 7s^2$

84. Aqueous solution of which metal cation is green in colour?

(1) Sc^{3+} (2) Ni^{2+}

(3) Zn^{2+} (4) Ti^{3+}

85. Bronze is a alloy of

(1) Cu + Ni

(2) Cu + Zn

(3) Cu + Sn

(4) Fe + Zn

86. Which among the following elements shows maximum number of oxidation states?

(1) Mn (2) Fe

(3) Co (4) Cr

87. Identify the correct increasing order of oxidising power among the following

(1) $MnO_4^- < Cr_2O_7^{2-} < VO_2^+$

(2) $VO_2^+ < MnO_4^- < Cr_2O_7^{2-}$

(3) $MnO_4^- < VO_2^+ < Cr_2O_7^{2-}$

(4) $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$

88. Which among the following has largest metallic radius?
(1) Ti (2) Sc
(3) V (4) Mn

89. Which of the following has highest number of unpaired d-electrons?
(1) Ni^{2+} (2) Cu^{2+}
(3) Fe^{2+} (4) Zn^{2+}

90. The hybridization involved in complex $[Co(C_2O_4)_3]^{3-}$ is (Atomic number of Co = 27)
(1) sp^3d^2 (2) d^2sp^3
(3) dsp^2 (4) sp^3d

BOTANY

91. Amongst the following, highest number of species is of
(1) Fishes (2) Beetles
(3) Ants (4) Orchids

92. The term biodiversity was popularised by
(1) Ernst Haeckel (2) Arthur Tansley
(3) Edward Wilson (4) Friederich

93. The diversity in the medicinal plant *Rouwolfia vomitoria* growing in different Himalayan ranges is an example of
(1) Genetic diversity
(2) Ecological diversity
(3) Environmental diversity
(4) Species diversity

94. Select the **incorrect** statement from the following.
(1) India has more than 50,000 genetically different strains of rice
(2) More than 1000 varieties of mango are found in India
(3) Ecosystem diversity in India is greater than that of Scandinavian country
(4) In India, Eastern Ghats have greater amphibian species diversity than the Western Ghats

95. According to IUCN (2004), the total number of plant and animal species described so far is
(1) About 30 million
(2) Slightly more than 1.5 million
(3) More than 15 million
(4) Less than one million

96. More conservative and scientifically sound estimate of global species diversity is made by
(1) Robert May (2) Jordan
(3) E.P. Odum (4) Norman Borlaug

97. Amongst the global species diversity, more than 70 percent are of
(1) Animals (2) Insects
(3) Fungi (4) Plants

98. The following pie chart is showing the global biodiversity of vertebrates:

Which sectors represent fishes and birds species respectively?
(1) E and B (2) B and D
(3) B and C (4) A and E

99. Which of the following representation regarding the proportionate number of species of major groups is **correct**?
(1) Ferns > Fungi
(2) Angiosperms = Gymnosperms
(3) Mosses < Algae
(4) Lichens > Algae

100. Out of every 10 animal species on earth, number of insect species is
(1) 5 (2) 2
(3) 4 (4) 7

101. India has only (i) percent of the World's land area, but shares (ii) percent of global species diversity.
Select the **correct** option to fill in the blanks (i) and (ii)
(1) (i) 4.5; (ii) 12.1 (2) (i) 2.4; (ii) 8.1
(3) (i) 8.7; (ii) 12.5 (4) (i) 1.7; (ii) 8.3

102. Identify the following statements as true (T) or false (F) and choose the option accordingly.

A. Our country is one among the 12 mega diversity countries of the world.

B. Nearly 95,000 species of plants have been recorded from India.

C. Biologists have estimated nearly all the species of prokaryotes present on earth.

	A	B	C
(1)	T	T	F
(2)	F	T	F
(3)	T	F	F
(4)	F	T	T

103. On earth, species diversity decreases as we move from

(1) Temperate regions towards tropical regions

(2) Higher latitude to lower latitude

(3) Equator toward poles

(4) East towards West on equator

104. Which of the following latitudinal range generally harbour more species than the others?

(1) 23.5° S to 47.0° S

(2) 23.5° N to 23.5° S

(3) 45.0° N to 90.0° N

(4) 23.5° N to 45.0° N

105. Regarding the number of birds species inhabiting an area, which of the following representation is **correct**?

(1) New York > India > Greenland

(2) Colombia > Greenland > India

(3) India > New York > Colombia

(4) Colombia > New York > Greenland

106. In South America, numbers of species belonging to which two groups is nearly the same?

(1) Mammals and amphibians

(2) Fishes and plants

(3) Birds and reptiles

(4) Amphibians and reptiles

107. For more species diversification a region should be

(1) More seasonal

(2) Less constant

(3) Relatively undisturbed

(4) At higher altitudes

108. On a logarithmic scale, the species-area relationship curve is

(1) J shaped (2) A straight line

(3) Hyperbolic (4) Bell shaped

109. In the equation log S = log C + Z log A that describes species-area relationship, Z represents

(1) Species richness

(2) Y - intercept

(3) Regression coefficient

(4) Area of that region

110. If we consider the species-area relationships for entire continents on a logarithmic scale, the slope of the line is found to be

(1) 0.6 to 1.2 (2) 1.8 to 2.0

(3) 2.5 to 2.8 (4) 0.1 to 0.2

111. Read the following statements and choose the option which is true for them.

Statement A: A stable community should not show too much variation in productivity from year to year.

Statement B: A stable community must be resistant to invasions by alien species.

(1) Both the statements are true

(2) Both the statements are false

(3) Only statement A is true

(4) Only statement B is true

112. By a long-term ecosystem experiments using outdoor plots, David Tilman found that

(a) Speciation is a function of time

(b) Plots with more species showed less year to year variation in total biomass

(c) Increased diversity contributed to higher productivity

The **correct** one(s) is/are

(1) (a) & (c) (2) (a) only

(3) (b) & (c) (4) (a) & (b)

113. In rivet popper hypothesis' proposed by Paul Ehrlich, an airplane is compared with

(1) An individual (2) Key species

(3) Species (4) Ecosystem

114. Match the following columns w.r.t. extinct species and the place where they were found.

Column-I	Column-II
(a) Steller's Sea Cow	(i) Africa
(b) Thylacine	(ii) Mauritius
(c) Dodo	(iii) Russia
(d) Quagga	(iv) Australia

	(a)	(b)	(c)	(d)
(1)	(iii)	(ii)	(iv)	(i)
(2)	(ii)	(i)	(iv)	(iii)
(3)	(iii)	(iv)	(ii)	(i)
(4)	(ii)	(iii)	(iv)	(i)

115. Presently, maximum percent of species of which of the following groups of organisms are facing the threat of extinction?

(1) Gymnosperms (2) Mammals

(3) Birds (4) Amphibians

116. Since the origin and diversification of life on earth, which extinction of living being is presently in progress?

(1) Seventh (2) Fifth

(3) Sixth (4) Fourth

117. Loss of biodiversity in a region does not lead to

(1) Increased speciation in a time period

(2) Decline in plant production

(3) Lowered resistance to environmental perturbation

(4) Increase variability in plant productivity

118. Amongst the following, which one is **not** the 'Evil Quartet' for biodiversity loss?

(1) Over-exploitation

(2) Glaciation

(3) Alien species invasions

(4) Co-extinction

119. Recently, the species that becomes a threat to the indigenous catfishes in our rivers is

(1) *Clarias batrachus*

(2) *Parthenium hysterophorus*

(3) *Pterophyllum scalare*

(4) *Clarias gariepinus*

120. All of the following are invasive weed species posing threat to our native species, **except**

(1) *Lantana* (2) *Eicchornia*

(3) Carrot grass (4) *Bryophyllum*

121. Total number of biodiversity hotspots in the world is

(1) 48 (2) 25

(3) 34 (4) 14

122. Which of the following is not an *ex situ* conservation strategy of organisms?

(1) Zoological park (2) Botanical garden

(3) National park (4) Wildlife safari park

123. To protect and improve the quality of our air, water and soil, in 1986, the Government of India has passed

(1) Environment (Protection) Act

(2) Forest Act

(3) Prevention and Control of Pollution Act

(4) Monetary Control Act

124. A scrubber is generally used to remove which of the following gases from exhaust?

(1) Sulphur dioxide (2) Carbon dioxide

(3) Chlorofluorocarbon (4) Carbon monoxide

125. One of the metallic elements, that can be used in automobiles to reduce the emission of poisonous gases is

(1) Copper (2) Rhodium

(3) Iron (4) Tungsten

126. The unit to measure the intensity of noise is

(1) Pascal (2) Lux

(3) Decibel (4) Watt

127. In all of the following aspects, CNG is different from petrol and diesel, **except**

(1) It burns most efficiently

(2) It cannot be siphoned off

(3) It cannot be adulterated

(4) It does not produce carbon dioxide

128. Regarding Euro III norms, in petrol, sulphur be controlled at

(1) 350 ppm (2) 50 ppm

(3) 150 ppm (4) 200 ppm

129. How many of the following is/are considered to be colloidal material in the waste water?

(1) Four (2) Three

(3) One (4) Two

130. By measuring the BOD of waste water we can estimate the amount of _____ present in that water.

Select the **correct** option to fill in the blank.

(1) Dissolved nitrogen gas

(2) Inorganic salts

(3) Biodegradable organic matter

(4) Carbon dioxide

131. The term 'Terror of Bengal' is related to

(1) An animal which is dangerous to human being in Bengal

(2) A plant which is the most problematic aquatic weed

(3) A viral disease very common in Bengal

(4) Water pollution caused by jute industries in Bengal

132. Increase in concentration of the toxicant at successive trophic levels is called

(1) Biomagnification (2) Biofortification

(3) Bioprospecting (4) Biodegradation

133. Which element in CFCs is mainly responsible to degrade ozone layer in the stratosphere?

(1) Carbon (2) Oxygen

(3) Fluorine (4) Chlorine

134. Chipko movement is related to protect the

(1) Wild animals from their hunting

(2) Trees from the axe

(3) Water from pollution

(4) Domestic animals from wild carnivores

135. One of the environment-friendly manner to treat the e-waste is

(1) Incineration

(2) Recycling

(3) Dumping

(4) Burying in the ground

ZOOLOGY

136. Which of the following set of animals have same level of body organisation?

(1) *Sycon, Hydra, Fasciola*

(2) *Ascaris*, *Pheretima*, *Pila*

(3) *Taenia*, *Ctenoplana*, *Nereis*

(4) *Wuchereria*, *Pheretima*, *Spongilla*

137. Select the **incorrect** match.

(1) *Obelia* – Metagenesis

(2) *Pleurobrachia* – Bioluminescence

(3) *Fasciola* – Dioecious

(4) *Hirudinaria* – Metameric segmentation

138. Which of the following features is/are common for both *Periplaneta* and *Pavo*?

(1) Presence of feathery wings

(2) Presence of scales

(3) Forelimbs are modified into wings

(4) Presence of crop and gizzard in alimentary canal

139. A pair of spermatheca is present in **A** abdominal segment of **B** cockroach which opens into the **C**. Choose the option which **correctly** represents A, B and C.

	A	B	C
1.	6th	Male	Genital chamber
2.	7th	Female	Genital pouch
3.	6th	Female	Genital chamber
4.	6th	Male	Vestibulum

140. Read the following statements **A** and **B**. Choose the **correct** option.

A. Matrix of cartilage is solid, pliable and resists compression

B. Matrix of bone is hard, nonpliable and rich in collagen fibres

(1) Both the statements **A** and **B** are correct
(2) Both the statements **A** and **B** are incorrect
(3) Only statement **B** is correct
(4) Only statement **A** is correct

141. Choose the option which **correctly** fills the blank to complete the analogy

PCT: Microvilli :: Fallopian tube____

(1) Brush border (2) Cilia
(3) Flagella (4) Villi

142. All of the following cells are present in loose connective tissue **except**

(1) Fibroblasts
(2) Mast cells
(3) Macrophages
(4) Chondrocytes

143. Nucleotides are formed when

(1) Phosphate group is found esterified to the sugar
(2) Sugar is found esterified with nitrogenous base
(3) Phosphate group is found esterified with heterocyclic rings
(4) Phosphate group is found esterified to sugar of nucleoside

144. Select the **incorrect** statement w.r.t. proteins present in our body.

(1) Proteins constitute maximum percentage of organic components of the total cellular mass
(2) Proteins are heteropolymers of different amino acids
(3) Receptor performs function such as sensory reception for smell, taste etc.
(4) GLUT-4 enables insulin transport into cells

145. The enzymes which catalyse oxidoreduction between two substrates are classified as

(1) Dehydrogenases
(2) Transferases
(3) Isomerases
(4) Hydrolases

146. Which of the following structures lining the villi produce numerous microscopic projections giving a brush border appearance?

(1) Cilia (2) Microvilli
(3) Villi (4) Cells

147. Select the **mismatch** w.r.t. cells of gastric glands and their secretions.

(1) Mucus necks cells – Mucus
(2) Oxyntic cells – HCl
(3) Parietal cells – Intrinsic factor
(4) Peptic cells – Renin

148. All of the following respiratory volumes/capacities represent vital capacity except

(1) TV + IRV + ERV
(2) IC + ERV
(3) TLC – RV
(4) TLC – FRC

149. Read the following statements w.r.t. respiratory gases and choose the option with all **correct** statement/s.

(a) pCO_2 in alveolar air is equal to PO_2 of blood present in pulmonary artery.
(b) Diffusion membrane for gaseous exchange is formed by three cellular layers
(c) Every 100 ml of oxygenated blood can deliver around 5 ml of O_2 to the alveoli under normal physiological conditions
(d) Chemosensitive area situated in rhythm centre is highly sensitive to pCO_2 and pO_2

(1) Only a
(2) b and c
(3) c and d
(4) a and d

150. Which of the following groups of animals circulate water in their body to provide nutrients and other useful substances as well as for removal of harmful substances from the body?

(1) *Sycon* and *Obelia*
(2) *Obelia* and *Nereis*
(3) *Asterias* and *Pheretima*
(4) *Taenia* and *Fasciola*

151. Select **mismatch** w.r.t. disorders of circulatory system.

(1) Hypertension – Blood pressure higher than normal
(2) Coronary artery disease – Narrower lumen of arteries due to

deposition of calcium, lipid and fibrous tissues

(3) Heart attack – Heart stops beating

(4) Heart failure – Heart is not pumping blood effectively enough to meet the needs of the body

152. Projections of renal pelvis are called

(1) Medullary pyramids

(2) Renal calyces

(3) Columns of bertini

(4) Ureter

153. Read the following statements and choose the option which represents all true and false statements **correctly**.

(A) Nearly 1% of glomerular filtrate formed per day is excreted in urine per minute

(B) Conditional reabsorption of Na^+ and water takes place in PCT

(C) Selective secretion of H^+, K^+ and NH_3 occurs in all segments of renal tubules except Loop of Henle

(D) About 1100-1200 ml of blood plasma is filtered by the kidneys per minute

	A	B	C	D
1.	F	F	T	F
2.	F	F	T	T
3.	T	T	F	F
4.	T	F	T	F

154. Choose the option which fills the blank correctly to complete analogy

HMM: Cross arm :: LMM : _____

(1) Tail

(2) Short arm

(3) Head

(4) Meromyosin

155. Total number of facial bones are equal to total number of bones in

(1) Right hand

(2) Thoracic cage

(3) Fingers of right hand

(4) Wrist of both hands

156. Match Column-I and Column-II and choose the option with all **correct** match

	Column-I		Column-II
a.	Myasthenia gravis	(i)	Increased chances of fracture due to decrease in bone mass
b.	Muscular dystrophy	(ii)	Autoimmune disorder
c.	Gout	(iii)	Genetic disorder
d.	Osteoporosis	(iv)	Inflammation of joints

1. a-(i), b-(ii), c-(iii), d-(iv)
2. a-(ii), b-(iii), c-(iv), d-(i)
3. a-(iv), b-(iii), c-(ii), d-(i)
4. a-(iii), b-(ii), c-(i), d-(iv)

157. Schwann cells are **not** found around

(1) Axon of myelinated nerve fibres of autonomic neural system

(2) Axon of unmyelinated nerve fibres of somatic neural system

(3) Axon of myelinated nerve fibres of PNS

(4) Axon terminals of myelinated nerve fibres

158. Select **mismatch** w.r.t. Human brain.

(1) Fore brain – Association areas and limbic lobe

(2) Cerebrum – Cerebral aqueduct, corpus callosum

(3) Hind brain – Pons, cerebellum and medulla oblongata

(4) Brain stem – Mid brain, pons and medulla

oblongata

159. Select the **incorrect** statement.
(1) Bony labyrinth is internally filled with perilymph
(2) Membranous labyrinth is surrounded by perilymph
(3) Perilymph filled bony labyrinth is divided into scala vestibuli and scala tympani
(4) Bending of hair cells generate nerve impulses in the associated efferent neurons

160. Which part of pituitary gland does not secrete MSH and is commonly called anterior pituitary?
(1) Pars intermedia
(2) Neurohypophysis
(3) Pars distalis
(4) Pars nervosa

161. All of the following are functions of glucocorticoids **except**
(1) Lipolysis
(2) Proteolysis
(3) Gluconeogenesis
(4) Increase uptake and utilisation of amino acids

162. Which of the following organisms have maximum life span?
(1) Butterfly (2) Crocodile
(3) Crow (4) Parrot

163. Read the following statements carefully and choose the option with only **correct** statements.
(a) Reproduction enables the continuity of the species
(b) Products of sexual reproduction which are genetically and morphologically similar are described as clones
(c) Formation of diploid zygote is universal in all organisms
(d) Zygote ensures continuity of species between organisms of one generation and the next
(1) a and b (2) b and c
(3) c and d (4) a and d

164. In human females, first meiotic division of primary oocyte is completed.
(1) Prior to ovulation
(2) Just after delivery of foetus
(3) During gestation
(4) During fertilization

165. Read the following statements A and B. Choose the **correct** option.
(A) Male germ cells are finally released from the seminiferous tubules by the process called spermiation.
(B) Fusion of male and female gametes form zygote which is implanted in uterus.
(1) Both the statements A and B are correct
(2) Both the statements A and B are incorrect
(3) Only statement A is incorrect
(4) Only statement B is incorrect

166. Choose the **correct** match w.r.t. gestation.
(1) Gestation period in Humans – About 9 months
(2) Appearance of hairs on the head – At the end of 2nd trimester
(3) Major organ systems are formed – At the end first month
(4) Mammary glands start producing milk – During 3rd trimester

167. All of the following are causes of population explosion in India **except**
(1) A rapid decline in death rate
(2) Increase in MMR
(3) Decrease in IMR
(4) Increase in number of reproductive individuals

168. Which of the following is **not** effective to be used as emergency contraceptive?
(1) Only progestogens
(2) Progestogen – oestrogen combinations
(3) Only oestrogen
(4) IUCDs within 72 hours of coitus

169. S.L. Miller created electric discharge in a closed flask containing gases
(1) CH_4, NH_3, H_2, H_2O
(2) CH_4, NH_3, N_2, H_2
(3) CH_4, NH_3, H_2 only
(4) CH_4, NH_3, N_2 only

170. Select the **incorrect** match.
(1) Charles Darwin – HMS Beagle
(2) Alfred Wallace – Malay Archipelago
(3) Karl Ernst von Baer – Approved proposal of Ernst Haeckel
(4) Lamarck – Theory of use and disuse of organs

171. Therapsids are direct ancestor of

(1) Mammals (2) Birds

(3) Pelycosaurs (4) Thecodonts

172. Read the following statements **A** and **B**. Choose the **correct** option.

Statement A: Common cold is one of the most infectious human ailments caused by Rhino viruses which infect lungs.

Statement B: Subsequent encounter with the same pathogen elicits a highly intensified amnestic response.

(1) Both the statements A and B are correct

(2) Only statement A is correct

(3) Only statement B is correct

(4) Both the statements A and B are incorrect

173. Widal test is a confirmatory test for

(1) Typhoid (2) Malaria

(3) AIDS (4) Hepatitis-B

174. Infective stage of *Plasmodium* for humans are **A** which are formed on **B** wall of mosquito and migrate to the **C**.

Choose the option which **correctly** represents A, B and C

	A	B	C
1.	Sporozoites	Mosquito	Liver
2.	Sporozoites	Gut	Salivary gland of mosquito
3.	Gametocytes	RBC	Gut of mosquito
4.	Merozoites	RBC	Salivary gland of human

175. Which of the following is **not** a physical agent to cause DNA damage leading to neoplastic transformation?

(1) X-rays

(2) Gamma rays

(3) UV light

(4) Tobacco smoke

176. All of the following are common side effects of anabolic steroids in both male and female **except**

(1) Increased aggressiveness

(2) Depression

(3) Excessive hair growth on the body

(4) Mood swings

177. Choose the option which fills the blank **correctly** to complete the analogy

Cross between different breeds: Cross-breeding :: Cross between closely related individuals of same breed : _____

(1) Outbreeding

(2) Outcrossing

(3) Inbreeding

(4) Interspecific hybridisation

178. Definition of biotechnology that encompasses both traditional view and modern molecular biotechnology is given by

(1) EFB (2) GEAC

(3) MOET (4) GMO

179. Read the following statements in which two statements are incorrect. Choose the option containing remaining **correct** statements.

(a) The normal *E. coli* cell carry genes encoding resistance against antibiotics ampicillin and tetracycline

(b) In plasmid pBR322, Insertional inactivation of tetracycline resistance gene enables transformants to grow in ampicillin containing medium

(c) In *E.coil* containing pBR322, non-recombinants can grow on the medium containing both ampicillin and tetracycline

(d) The recombinants of *E.coli* with pBR322, having insert at *Bam*HI, will grow in medium containing tetracycline

(1) a and b (2) b and c

(3) c and d (4) a and d

180. Select the **incorrect** match.

(1) Emphysema – α – 1–antitrypsin

(2) Rosie milk – Lactose-enriched

(3) ELISA – Antigen - antibody interaction

(4) PCR detect – Routinely used to HIV infection

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